SAT and SMT solvers, WDSat and Nagao decomposition for the PDP

SAT and SMT solvers, WDSat and Nagao decomposition for the point decomposition problem: what is measured, what the literature adds, what is left

Date. 2026-10-10. **Status.** Survey and gap analysis. Nothing new is measured here; every number below is quoted from a frozen repository artefact or a cited paper, and each row says which. **Class.** Accounting (no operation count moves). **Conductor task.** T-1. **Visuals.** research/pdp_sat_smt_survey_20261010/ (oracle_ladder.svg, product_law.svg, REPORT.pdf).

The question. Is there anything in SAT or SMT solving for the point decomposition problem (PDP) on binary curves that this repository has not exhausted and that could give a better speed-up, and is Nagao's decomposition a better route than summation polynomials? The prompt behind it is the Trimoska–Ionica–Dequen WDSat line, which reports large speed-ups over Gröbner bases on exactly these systems.

Bottom line.

1. **As an exponent lead the SAT route is exhausted, and the repository has measured why.** On the Weil-descended Semaev system with the summands in an F_2-subspace V of dimension ℓ, every SAT, SMT and WDSat back end tried here spends about one conflict per candidate tuple of the factor base. It is an enumerator: plain WDSat visits $2^{3\ell}/3!$ nodes to within 0.2 %, the native CDCL+XOR solver 0.97–1.02 conflicts per triple from $\ell = 5$ to 8, CryptoMiniSat $36.7\times$ the pair count at $d = 8$ on the ECC2K-130 curve. The best oracle in the tree enumerates $m - 1$ summands and root-finds the last ($2^{(m-1)\ell}$, with memory $2^{\lceil m/2 \rceil \ell}$); it beats every solver by 10^2–10^4 at the largest rung both finish (§2). WDSat's published speed-ups are real and are relative to Magma F4, which on these systems is worse than enumeration (deciding degree $\approx \ell/2$, §3.2); they are not speed-ups relative to enumeration.
2. **No solver constant matters for ECC2K-130.** The product law relations × targets × oracle = $m \cdot 2^n$ ($/n$ for a Frobenius-stable base) puts any oracle that enumerates sub-tuples $2^{+64\ldots72}$ above rho; an oracle would have to beat exhaustive search over its own candidate set by $2^{-(70.19 + \log_2 m - \log_2 131)}$ (§1). A SAT solver cannot do that, and neither can an SMT solver, whose Boolean and bit-vector theories bit-blast to the same CDCL core (measured: cvc5 and Z3 within 0.44–$2.6\times$ of the native solver at $n = 13$, §2).
3. **Nagao's decomposition is not a better route at prime extension degree.** Nagao's gain is a *composite-degree* phenomenon: with a subfield factor base the condition "the norm polynomial has coefficients in F_q" is a quadratic Weil-restriction system with no summation polynomial to expand. At prime $n = 131$ there is no subfield, the condition becomes "roots in V", i.e. divisibility by the degree-2^ℓ subspace polynomial, and the repository's Riemann–Roch encodings reduce to S'_4 with the linear part hidden. Measured on ECC2K-130: the RR solver resolves $32/32$ matched slots where the Semaev controls resolve $0/32$, and brute-force pair enumeration then costs $0.494\times$ the RR solver's counted field operations (§5).
4. **What is not exhausted is engineering, and it is listed with falsifiers in §6.** The cheapest items reuse harnesses already in the tree: a matched run of the newer SATIC ANF solver and an audit of WDSat's incomplete `-x` mode on the frozen 60-input regression; a PB-XOR solver on the one factor base where SAT has a structural edge over Gröbner (Hamming-weight in a normal basis); ANF preprocessing (Bosphorus) in front of every SAT arm; and completing the SMT ladder to $n = 23$ so the open SMT study can be classed. None is expected to move a ratio to the floor; each is a stage diagnostic.

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1. Boundary, stated before anything else

Setting: `E/F_{2^n}`, `n` prime (ECC2K-130: `K₀: y² + xy = x³ + 1`, `n = 131`, `#E = 4r`, `r ≈ 2^129`), factor base `F = {P : x(P) ∈ V}` with `V` an `F₂`-subspace of dimension `ℓ`, or a Frobenius-stable set (Hamming-weight in a normal basis), `m` summands.

Reference. Pollard rho with the `⟨−1⟩ × ⟨π⟩` automorphism discount: `2^{60.81}` group operations on ECC2K-130 (RESEARCH_ECC2K130_DECOMPOSITION.md §5.2).

Floor for the whole method (the product law, same note §5.1).

```
   relations · targets/relation · oracle/target
 = 2^ℓ · 2^n / C(2^ℓ, m) · C(2^ℓ, m−1)  ≈  m · 2^n          (independent of ℓ)
```

divided by `n` when the base is Frobenius-stable. Turned around (§5.3 there): an oracle that reaches rho must beat exhaustive search over its own candidate set by `2^{−(70.19 + log₂ m)}`, or `2^{−(63.2 + log₂ m)}` with the Frobenius collapse. No tuning of `m`, `ℓ` or the solver changes that number; it is the identity. The free-oracle floor (charge nothing per decision) sits below rho from `m = 4` on (`2^{56.40}`), so the method is not structurally impossible — it is the oracle that protects the curve.

Floor for the per-target oracle (this note's unit). The trivial oracle enumerates `m − 1` summands and decides the last by root-finding the univariate `S_{m+1}` against `V`; with the `S_m` symmetry that is

```
   T_enum(m, ℓ) = 2^{(m−1)ℓ} / (m−1)!    candidate sub-tuples per target,
```

each costing `O(ℓ)` field operations through `gcd(q, L_V mod q)` (RESEARCH_SEMAEV_DECOMPOSITION.md). With `2^{⌈m/2⌉ℓ}` memory a pair table (meet in the middle) lowers it to `2^{⌊m/2⌋ℓ}` lookups. A solver that reports more than `T_enum` conflicts or decisions per target is enumerating with overhead; one that reports fewer is doing algebra the enumerator does not. **Unit:** the ratio of a solver's per-target work to `T_enum`, in the solver's own counter where the artefact reports one and in counted field operations where it does not. Wall time is quoted beside it as a practicality note only.

Falsification target for the "exhausted" verdict. The verdict is wrong if any solver, encoding or preprocessing in §6 produces, on uniform (not planted) targets at `m = 3`, a per-target count below `0.5 · T_enum` at two consecutive rungs `ℓ ≥ 8`, with every verdict checked against the exhaustive oracle. It is confirmed if every arm stays at or above `T_enum` out to the largest rung it decides.

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2. The oracle ladder: one table

Every row is a frozen artefact; the right-hand columns are the ratio to the enumeration floor of §1 on the largest rung the arm decides, and the verdict the artefact records. `m = 3` unless stated. "Planted" rows measure first-solution cost on a target known to decompose; "uniform" rows measure exhaustion cost, which is the oracle's real job.

arm	encoding → solver	base	largest rung decided	work per target vs `T_enum`	correctness	verdict in source	source
pairs-and-solve (reference)	S_4 quartic in last variable, `gcd(q, L_V mod q)`	subspace	$\ell = 12$, $n = 36$ (11.5 G triples in 15 s)	`1.0×` by construction; `8 µs → 10 ns` per candidate after one-word field	exhaustive, cross-checked	the oracle the ledger prices	RESEARCH_SEMAEV_DECOMPOSITION.md, RESEARCH_SAT_SEMAEV.md §"What was done instead"
direct MITM / pair table	group law, no polynomial	subspace, Frobenius orbits	$n = 59$, $\ell = 9$ (80/80 verdicts, 3.86 core-s median)	$2^{-\ell}\times$ time at $2^{2\ell}$ memory	160/160	"lowest median core time in every cell"	research/sat_factor_base_review_20260908/…/STAGE13_RESULTS.md, STAGE26_32_PHASE_B_RESULTS.md
matrix-F4 with splitting	Weil-descended S_3 chain / S_4, Macaulay rows	subspace	$n = 21$, $m = 3$ refuted in 50 s; $n = 31$, $m = 3$ refutation 145 s	$8.8\cdot10^4$ GAE at $n = 23$, $m = 2$ against 0.011–25 GAE for the pair table	agrees with enumeration target by target	best *algebraic* oracle; deciding degree $\approx \ell/2$	RESEARCH_IC_BOUNDARY_LEDGER.md §5, RESEARCH_KOBLITZ_SCALING_TARGET.md
native CDCL + Gauss–Jordan XOR (`sat.rs`)	symmetrised S_4 → native parity rows, free-variable branching, lex symmetry breaking	subspace	$n = 24$, $\ell = 8$ uniform rejection: 2,877,905 conflicts on 2,829,056 triples	≈ 1.02 conflicts per *triple*, i.e. $\approx 2^{\ell}/3 \times$ `T_enum`; wall 62× brute force and widening (5×, 8×, 18×, 62× over $\ell = 5$…8)	all verdicts match enumeration	"SAT is a strictly worse way to enumerate the factor base"	RESEARCH_SAT_SEMAEV.md §"Does it scale? No."
native CDCL + XOR, IC pipeline (`sat_decompose`)	chained S_3 + Kosters–Yeo trace row + degree-2 Macaulay rows	invariant subspace	$n = 15$, $m = 3$ refuted in 25.3 s (F4: 75 ms); $n = 21$, $m = 3$ no verdict in 31 min	$1.3\cdot10^7$ GAE at $n = 23$, $m = 2$ (150× F4)	agrees where it answers	fails 2/16 targets at 20 unknowns, 10/16 at 22	RESEARCH_KOBLITZ_SCALING_TARGET.md §240–305, ledger §17.11
WDSat `61c6ff3`, plain and symmetry-broken	Trimoska ANF (`.anf`), XORGAUSS off	subspace	frozen corpus $n \in \{15, 17, 19\}$, $\ell \in \{5, 6\}$, 60 inputs	symmetry on/off: 1,280,281 / 6,246,481 summed median conflicts (4.88×, the $3!$ of S_3)	30 SAT, 29 UNSAT, 1 algebraic-only rejection, all certified	regression target, "no solver algorithm was changed"	research/index_calculus_baseline_20260914/regression/RESULTS.md
WDSat, Semaev ANF, $n = 59$, ℓ	same	subspace	none: outer watchdog at	264× direct-MITM wall, no	—	inconclusive	…/continuation-05-sota-gates/GATE_STATUS.md Stage 193

arm	encoding → solver	base	largest rung decided	work per target vs T_enum	correctness	verdict in source	source
= 9			120 s on all 40 targets	terminal			
WDSat on the real curve (solver_16)	S'_4, polynomial basis, V = {deg x < d}	subspace, ECC2K-130	d = 8	plain: 2^{3d}/3! decisions to 0.2 % (= 2^d/3 × T_enum); -x returned UNSAT on a decomposable target in 1 of 16 cells	-x **incomplete** at n = 131	"plain WDSat is brute force"	RESEARCH_ECC2K130_RR_SOLVER_PANEL.md §11
CryptoMiniSat 5.14 on the real curve (solver_16)	S'_4 → CNF + native XOR	subspace, ECC2K-130	d = 8 (d = 9, 10 censored at 300 s)	uniform exhaustion: 1 conflict at d ≤ 6 (the linear certificate), 839 at d = 7 (0.31× pairs), 312,231 at d = 8 (36.7× pairs)	every witness verified	"super-quadratic in |F| from d = 7"	same, §11
CryptoMiniSat on the ONB Karatsuba circuit	gate-level CNF + XOR, Sinz cardinality for the weight bound	**Hamming-weight w = 3** in the normal basis (Frobenius-stable at prime n)	n = 11 (median 2.64 s); n = 23 nothing in 25 min	≈ 2^{2.3} per field bit; encoding untuned	planted	the only SAT-only base; "the shape is already clear enough"	ecc2k130/README.md §Measured
Kissat 4.0.4 / CaDiCaL 3.0.1	Tseitin CNF of the five-summand S_3 chain	window slots, n = 83	none with the target free (120 s); half-pinned planted instance 157,941 conflicts in 30 s, no model	—	planted control only	inconclusive	research/f6_n83_factored_s3_sat_20261005/RESULT.md
cvc5 1.4.2 / Z3 (QF_UF, QF_BV)	identical rows to the native arm, SMT-LIB 2	subspace	n = 13, ℓ = 5 (20/20 each); n = 19 every	Z3-bool 0.76×, Z3-bv1 1.82× native conflicts; cvc5-bool	100/100	unclassed: parity reasoning is	research/pdp_smt_oracle_20261008/README.md

arm	encoding → solver	base	largest rung decided	work per target vs `T_enum`	correctness	verdict in source	source
			arm at the 120 s cap	`0.44×`, Z3 `2.2–2.6×` native wall		not the gap; `n = 23` not run	
Crossbred (`D = 2, 4`)	Macaulay extraction + `2^k` linear solves	subspace	`n = 9, m = 3`	`xb/F4 = 0.696` wall and rising; **zero filters at any `(D, k)`**; pipeline `1800×` behind enumeration	agree	"enumeration in disguise"	RESEARCH_ECC2K130_CROSSBRED.md
Riemann–Roch / Nagao (`quadratic-image`)	function coefficients `(a, b)` as unknowns, norm-polynomial coefficient match, support test `H | L_V`	subspace, ECC2K-130	`d = 6, 7` (32/32 matched slots; 28 verified 3-summand relations on the real curve)	`0(|F|²)` candidates; pair enumeration `0.494×` its counted field ops	every witness re-derived in the group	"the null object wins"	RESEARCH_ECC2K130_RR_SOLVER_PANEL.md §7–8
Riemann–Roch under SAT (WDSat / CryptoMiniSat)	RR-norm and RR-incidence ANF	subspace	RR-norm `n = 9` (8/8); RR-incidence does not fit WDSat (SIGSEGV above `≈ 23` M table entries)	XORGAUSS `79,151 → 62` conflicts (`1,277×`); RR costs `3.5·10⁴×` the S'_4 conflicts at `d = 5`	witnesses verified	"the RR encoding is S'_4 with the linear part hidden"	RESEARCH_ECC2K130_WDSAT.md, panel §11
Hamming-ideal Gröbner (La Scala–Marchesin–Tiwari lifts)	S_3 + C / FC / QFC Hamming ideals, `MultiSolve`	Hamming-weight, normal basis, `m = 2`	`n = 19`	calls per target grow as `|F_w|^{1.9–2.5}` against a target exponent `0.8`; tame depth within 1–3 of a full summand	all targets correct	"the exhaustive oracle with a truncated Gröbner computation attached"	RESEARCH_HAMMING_IDEAL_PDP.md
one-hot Frobenius-phase SAT	orbit representative + shift variables	Frobenius orbits	`n = 13` (`≈ 50 s` median); `n = 19` fails at 1,200 s	—	—	no gain	`research/frobenius_quotient_sat_20260930/`

Three readings.

- **Every arm above the reference line is at or above `T_enum`.** The two that look below it — CryptoMiniSat at $d \le 7$ on the real curve, and matrix-F4 at $m = 2$ — are the linear NO-certificate of the S_4 value set (available only for $d \le 6$, saturating at $d = 7$; panel §9) and the Kosters–Yeo trace fall at $m = 2$ (first fall degree 2, a theorem), which the enumerator also gets for free.
- **The SAT arms sit a factor $2^{\ell/m}$ *above* `T_enum`, not at it.** One conflict per *full* tuple means the solver assigns all m summands before the descended equations bind; the enumerator assigns $m - 1$ and root-finds. Only WDSat's `-x` mode might close that gap, and the one artefact that tests it at full size records a wrong UNSAT (§6, item 1).
- **Gröbner is the arm WDSat beats, and it is below enumeration too.** The deciding degree of F4 on the one-summand-fixed system runs `5, 6, 6, 7, 7` over $\ell = 3\ldots7$ (slope `0.500`), so elimination alone is $\approx 2^{3.6\ell}$ against $2^{2\ell}$ for the enumerator (decomposition note §5.4). That is the baseline the WDSat papers improve on.

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3. Why every solver is enumeration in disguise

This is the structural reason the §6 list is engineering and not a new lead. It is a derivation from measured fall degrees, not a theorem.

3.1 Where the equations bind

Weil descent of $S_{m+1}(x_1, \ldots, x_m, x_R)$ with $x_i \in V$ gives n Boolean equations in $m\ell$ unknowns. In characteristic 2 the even powers x^{2^k} are F_2-linear in the coordinates, so the descended system is quadratic for $m = 2$ and cubic when S_3 is chained ($m \ge 3$); the symmetrised S_4 is quadratic in the elementary symmetric unknowns with a cubic correspondence system behind it. With $m - 1$ summands fixed the residual is a univariate polynomial of degree 2^{m-1} in the last x, whose roots in V are found by `gcd` with L_V in $O(\ell)$ field operations: that is the enumerator's leaf. With only $m - 2$ fixed, the residual is the $m = 2$ problem for the target $R - P_1 - \ldots$: n quadratic equations in 2ℓ unknowns.

3.2 The $m = 2$ residual is not cheaper than 2^{ℓ} by any known method

- **Enumeration:** 2^{ℓ} candidates for x_2, each a group subtraction and a membership test. This is what pairs-and-solve and the pair table do.
- **Gröbner/XL on the residual:** the first fall degree is 2 (Kosters–Yeo Cor. 4.11, the trace row), but that fall determines one bit; the measured deciding degree grows as $\approx \ell/2$ and the only proved upper bounds on the last fall degree are linear in n (RESEARCH_FALL_DEGREE_BOUNDS.md §2.3). Kosters–Yeo also show that "$d_{reg} \approx d_{ff}$" for the split system would decide an NP-complete problem in polynomial time, so the subexponential claims that rest on it (Petit–Quisquater, Semaev 2015, Kousidis–Wiemers) are unproved heuristics, and the toy data here go the other way.
- **Generic MQ solvers on the residual:** $N = 2\ell$ unknowns and $n \approx 1.5\,N$ equations. The best generic exponents over F_2 (Dinur's polynomial method $2^{0.6943\,N}$ for $M = N$, Crossbred and FES around $2^{0.8\,N}$, Lokshtanov et al. $2^{0.8765\,N}$) give at best $2^{1.39\,\ell}$, above the 2^{ℓ} enumerator. Crossbred measured on these systems found no filters at any (D, k) and a margin over F4 that shrinks with n.
- **CDCL/DPLL on the residual:** the quadratic rows do not propagate until nearly all of x_2's bits are assigned, which is the "tame depth within 1–3 of a full summand" the Hamming-ideal thread measured and the one conflict per leaf the SAT threads measured.

So the leaf cost of any tree search on this system is bounded below by the $m = 2$ residual, and the $m = 2$ residual is bounded below by 2^{ℓ} on all measured and all proved evidence. A solver can only lose constants against T_{enum}; the table says the measured ones lose $2^{\ell}/m$.

3.3 What a solver *can* do that the enumerator cannot

Two things, both already in the tree:

- **Cardinality constraints.** A Hamming-weight factor base in a normal basis is Frobenius-stable at prime n and has a popcount membership test; its membership ideal has degree ≥ 3 and $O(n \log n)$ auxiliaries (Hamming ideals), which Gröbner handles badly and a SAT solver handles natively. This is the one base where the solver family matters. The product law is unchanged by the base, so the gain is confined to the oracle constant.
- **UNSAT certificates.** CaDiCaL with DRAT-trim produced externally checked refutations of five $n = 13$ branch queries (`research/notes/ecc2k130/n13_oaware_unsat_proof_20260925/`). That is a correctness instrument, not a speed.

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4. The literature, 2019–2026

4.1 SAT and SMT solvers on the descended systems

Sources were read in full where the sandbox could fetch them (Trimoska's thesis, the CP 2020 and SAT 2021 papers, GGMP 2020, Ozdemir et al. CAV 2023, Hader–Ozdemir 2024, Danner–Kreuzer JAIR 2026); the items that could only be read as abstracts are marked so.

WDSat (Trimoska–Ionica–Dequen; AFRICACRYPT 2020, CP 2020, SAT 2021; thesis 2021). The solver is a DPLL (no clause learning) with three propagation modules: CNF unit propagation, parity propagation (XORSET) and Gaussian elimination on the parity system with substitution tracking read straight from ANF (XORGAUSS, "XG-ext"). It branches only on the $m \cdot \ell$ coordinate bits, and breaks the S_m symmetry inside the search by refusing the branch that would violate $x_1 \leq \cdots \leq x_m$ (gain exactly $m!$, no overhead). Two facts from the thesis decide this note's question:

- **Rigorous bound.** Conflicts per PDP are at most $2^{m\ell}/m!$ (thesis eq. 7.14), and the relation-search phase for S'_4 is $\tilde{O}(2^{n+\ell})$ (Thm 7.5.1), against rho's $2^{n/2}$. The authors say so themselves.
- **The bound is attained.** Thesis Table 7.6, UNSAT instances, $m = 3$: $2.80 \cdot 10^6$ conflicts at $\ell = 8$ against $2^{24}/6 = 2.796 \cdot 10^6$; $22.39 \cdot 10^6$ at $\ell = 9$ against $22.37 \cdot 10^6$; $179.0 \cdot 10^6$ at $\ell = 10$; $1.432 \cdot 10^9$ at $\ell = 11$ against $1.431 \cdot 10^9$. WDSat prunes nothing above the last branching level ("the system generally does not become linear until the second-to-last branching"). The per-candidate cost at $\ell = 11,\ n = 89$ is $\approx 176{,}000\ \text{s} / 1.43 \cdot 10^9 \approx 120\ \mu\text{s}$, four orders of magnitude above this repository's one-word enumerator.

Measured on Koblitz curves (one thread, Xeon E5-2640): Magma F4 takes 207 s at $\ell = 6,\ n = 17$ (3.6 GB), 3,855 s at $\ell = 7,\ n = 19$ (38 GB) and runs out of 200 GB at $\ell = 8$; WDSat with symmetry breaking takes 0.22–0.61 s, 2.2–6.9 s and 30–86 s on the same rungs, and reaches $\ell = 11,\ n = 59 \ldots 89$ in 1.5–2 days per instance in 19 MB. The headline "up to 1700× over Gröbner" is this comparison at $\ell = 6, 7$. Generic CDCL solvers (MiniSat, Glucose, CaDiCaL, CryptoMiniSat with the authors' branching patch) stop at $\ell = 8$. For S_3 ($m = 2$) the XG-ext module with a minimum-vertex-cover branching order solves $n = 41,\ \ell = 20$ in 4–14 s; the MVC has $\ell = 20$ elements, i.e. the solver guesses one summand and solves a linear system, which is the trivial $m = 2$ oracle. Gaussian elimination *hurts* on S'_4 (the monomial graph is

complete, so the MVC is every variable). The thesis' own reality check: at $n = 59$ their parallel collision search takes 0.8 h single-threaded, while 2^9 successful decompositions at $\ell = 9$ would take more than 86 h.

Follow-ups. Nothing after 2021 re-ran WDSat on PDP or passed its $\ell = 11$ instances (Semantic Scholar citation graph, checked 2026-10). The direct successor is Blomme, Cherif, Ionica, Dequen, *ANF-based satisfiability for Weil-descent cryptographic attacks* (CoDIT 2025, HAL hal-05176414, abstract only: lazy watched structures on the ANF itself, no CNF-XOR translation); its benchmark sizes are unverified. Andraschko–Danner–Kreuzer (2-XNF, Math. Comput. Sci. 2024) and Danner–Kreuzer (CDXCL / Xorricane, JAIR 2026) add clause learning to XOR-native solving with a proof system (XLIN) polynomially equivalent to resolution with parity, which is exponentially stronger than resolution; it was benchmarked on Ascon, Bivium, CTC2 and random MQ, not on PDP, and loses to Kissat on structured instances. CryptoMiniSat ≥ 5.8 (Soos–Gocht–Meel, CAV 2020) runs bit-packed Gauss–Jordan at every decision level; Trimoska's CMS numbers predate it, and this repository's CMS 5.14 runs are the post-5.8 data point (§2: $36.7\times$ pairs at $d = 8$). Bosphorus (Choo–Soos–Chai–Meel, DATE 2019; ANF/CNF simplification with XL, ElimLin and learnt facts) is untested on Weil-descent systems anywhere. Cube-and-conquer in cryptanalysis exists only for hash inversion (Zaikin, JAIR 2022). ML-guided SAT benchmarks (SAT4CryptoBench, NeurIPS 2025) carry no ECC instances. In-memory XOR–CNF hardware (Im et al., Nat. Commun. 2026) and GPU clause evaluation (GaloisSAT, 2026) report simulated or generic-instance gains and cite WDSat without running PDP. No hybrid Gröbner+SAT or XL+SAT paper on PDP appeared 2021–2026.

What the SAT literature never does is compare against enumeration. Every paper's baseline is F4; the MQ literature's baseline is FES (Bouillaguet et al., CHES 2010: Gröbner $\approx 10^6\times$ slower than libFES on F_2; Joux–Vitse Crossbred 2017; Dinur 2021 $2^{0.6943\ N}$; Vidal–Delaplace–Ionica, IACR CiC 2025 on Crossbred). This repository made the comparison (§2) and the answer is the one the thesis' Table 7.6 already implies.

SMT. cvc5's finite-field theory (Ozdemir–Kremer–Tinelli–Barrett, CAV 2023; split Gröbner bases, CAV 2024) is a Gröbner-basis decision procedure for **prime** fields; the SMT-LIB theory proposal (Hader–Ozdemir, SMT 2024) defines extension sorts `(_ FiniteField p n)` but no solver implements them (cvc5 and Yices support `QF_FFA` for prime fields only). Bit-vector encodings of F_{2^n} arithmetic are a Weil descent with a Tseitin variable per multiplication gate, 10–20× larger than the descended ANF (thesis Table 7.1), and bit-blast to the same CDCL cores (cvc5 → CaDiCaL, Z3 → its own); Ozdemir et al. report bit-vector solvers already inferior on fields of $\geq 2^{40}$ elements. No peer-reviewed paper evaluates any SMT solver on ECDLP, PDP, `GF(2^n)` multiplication constraints or Semaev polynomials. This repository's own SMT study (§2) is therefore the only data: cvc5 and Z3 land within 0.44–$2.6\times$ of the native solver at $n = 13$ and hit the cap with it at $n = 19$. A native `GF(2^n)` theory would be Gröbner bases again. Verdict: no unexploited SMT avenue.

Koblitz specifics (Galbraith–Gebregiyorgis–Murphy–Petit, SAC 2020). The $m!$ symmetry can be put in the *model* with π-shifted factor bases $F_i = \pi^{i-1}(F)$ and recovered as $R = P'_1 + \lambda P'_2 + \cdots$; it is the same $m!$ WDSat takes in the search, so the two do not stack, but GGMP's form works under any solver (F4, FES). Frobenius-invariant subspaces need a factor of $x^n - 1$; at $n = 131$ it is $(x - 1)\cdot\Phi_{130}$ with $\mathrm{ord}_{131}(2) = 130$, so the only invariant subspaces are $\{0\}$, F_2, the trace-zero hyperplane and the field (this repository's §6 of the decomposition note, independently). GGMP's conclusion: "our improvements do not lead to index calculus faster than Pollard rho on instances used in practice."

Records. End-to-end binary ECDLP by Semaev index calculus: $n = 19$ (Huang–Petit–Shinohara–Takagi 2013); relation collection only at $n = 24$ (WDSat, $\ell = 8$, 18 h). PDP by Gröbner: $n = 53$, $\ell = 6$ symmetrised S_4 (HPST, 35 s); $m = 2$ at $n = 45$ needed 126 GB with $D_{reg} = 5$ (Kosters–Yeo). PDP by SAT: $\ell = 11$, $n = 89$ ($m = 3$, WDSat); $\ell = 20$, $n = 41$ ($m = 2$). This repository's panels sit inside those: $n = 59$, $\ell = 9$ by MITM in 3.9 core-s where WDSat and CMS time out, and $\ell = 12$ by pairs-and-solve in 15 s. Every binary ECDLP record remains a rho record; ECC2K-130 is unsolved and "no known index calculus attack applies" (Bailey et al. 2009) is still the state of the art.

4.2 Nagao's decomposition and the other un-eliminations of S_{m+1}

Nagao (ePrint 2007/112; ANTS 2010). For $\mathrm{Jac}(C)/\mathbb{F}_{q^n}$ with factor base "$x \in \mathbb{F}_q$", decompose a divisor D by finding $f \in L(ng\cdot O - D)$ whose norm polynomial $F(x) = f(x, y)\, f(x, -y)/u(x)$ has **all coefficients in $\mathbb{F}_q$**: the coefficients are quadratic in the unknown coefficients of f, and the $\mathbb{F}_q$-rationality condition Weil-restricts to $(n-1)\cdot ng$ quadratic equations in $(n-1)\cdot ng$ unknowns over $\mathbb{F}_q$, generically zero-dimensional of degree $2^{(n-1)ng}$. For $g = 1$ that is $n(n-1)$ quadrics in $n(n-1)$ unknowns against Semaev's n equations of degree 2^{n-1} in n symmetric unknowns — **the same Bézout degree $2^{n(n-1)}$, un-eliminated**. No first-fall or D_{reg} data for the elliptic case is published. Nagao 2013/549 claims a subexponential bound under a first-fall-degree assumption of the kind Kosters–Yeo and Huang–Kosters–Yeo later refuted experimentally.

Joux–Vitse (J. Cryptology 2013; EUROCRYPT 2012). The people who built both: "This approach is less efficient than Semaev's in the elliptic case, but is the simplest otherwise [higher genus]"; Nagao-style tests are feasible for $(n, g) \in \{(2, 2), (2, 3), (3, 2)\}$ and cost 22 ms per core on their genus-3 cover over $\mathbb{F}_{p^2}$, where their own sieve on $F(x)$ is "about 960 times faster". Their elliptic record ($E(\mathbb{F}_{p^5})$, 130-bit, oracle-assisted static DH) uses Semaev with the $n-1$ trick (n equations, $n-1$ unknowns, degree 2^{n-2}) and F4 trace replay, not Nagao. Faugère–Huot–Joux–Renault–Vitse (EUROCRYPT 2014) then take S_6 symmetrised under 2-torsion to 10 s per decomposition on the Oakley curve over $\mathbb{F}_{2^{155}} = \mathbb{F}_{2^{31}}^5$ ("too slow to seriously threaten the DLP"). Galbraith–Gaudry (DCC 2016): Nagao's approach "is also valid for elliptic curves … nobody managed to turn this into an algorithm that is faster than an approach based on summation polynomials."

The prime-degree obstruction. Nagao's and Gaudry's formulations both presuppose a factor base over a proper subfield $\mathbb{F}_q$, q large. Over $\mathbb{F}_{2^{131}}$ the only subfield is $\mathbb{F}_2$, so the rationality condition has nothing to act on. The replacement, "all roots of F lie in the subspace V", is divisibility by the linearised polynomial L_V of degree 2^ℓ, which is not coefficient-linear; writing it out either re-introduces a root variable per summand (and lands on the FPPR / HPST symmetrised system) or keeps the coefficients as unknowns and tests $H \mid L_V$ by iterated squaring — which is this repository's function-first encoding (`nagaoannihilator`, `solver_02…09`), exhaustive in $q(q-1)$ coefficient pairs and $\theta(|F|^2)$ in its best image-space form (§2). Galbraith (ECC 2015) states the real question: "Semaev's approach is to minimize number of variables at the expense of exponential degree. Other choices of coordinates can lead to lower degree but more variables. Problem: determine the optimal tradeoff." The published points on that axis are FPPR's symmetrised descent, HPST's mixed symmetric/non-symmetric variables (D_{reg} 6–7 → 3–4; whole ECDLP at $n = 19,\ m = 3$ in 9,963 s), Semaev's 2015 S_3 chains (cubic Boolean system, first fall 4 proved, $D_{reg} = 5$ already at $n = 45$ by Kosters–Yeo), Karabina's auxiliary-variable chains ($m = 3$: $n = 26$ in 1,646 s; $m = 4,\ 5$: $n = 19$; an explicitly exponential $2^{34}\cdot 2^{n/5}\cdot n^{27}$ extrapolation), and Galbraith–Gebregiyorgis' binary-Edwards T_2-invariant coordinate ($D_{reg} \le 4$ at $m = 3$ out to $n = 97$, $\ell \le 7$; $m = 4$ only $\ell \le 4$ under Gröbner, $\ell \le 7$ under MiniSat at 4 % success). This repository has measured two of those points (chained S_3: cubic, slower in every engine, §2; symmetrised S_4 under SAT and F4) and the T_2 symmetrisation on K_0.

Koblitz structure. Galbraith–Granger–Merz–Petit (SAC 2020): Frobenius $\pi = [\lambda]$ breaks the $m!$ symmetry in the model and a Galois-invariant subspace V_f (roots of a linearised polynomial from an irreducible factor $f \mid x^n - 1$) divides relation collection by n and linear algebra by n^2. The construction needs $\deg f = \mathrm{ord}_n(2)$ useful; checked here by direct computation: $\mathrm{ord}_{127}(2) = 7$, $\mathrm{ord}_{131}(2) = 130$, $\mathrm{ord}_{163}(2) = 162$, $\mathrm{ord}_{233}(2) = 29$, $\mathrm{ord}_{283}(2) = 94$, $\mathrm{ord}_{571}(2) = 114$. So at $n = 131$ the invariant subspaces are $\{0\}$, $\mathbb{F}_2$, the trace-zero hyperplane and the field (the decomposition note's §6 and `ecc2k130/README.md` say the same), while K-283 and K-571 happen to have invariant subspaces of dimension $\approx n/3$ and $\approx n/5$ — a structural curiosity, far outside any reachable PDP size, noted and not pursued. Tsakou–Ionica (ePrint 2021/721) extend endomorphism-reduced bases to composite-degree Jacobians. Nothing on prime-degree binary PDP appeared 2022–2026 in ePrint or arXiv; Courtois' $2^{n/3}$ splitting claims (ePrint 2016/003) have no independent check and the author's own text says they may "violate the generic group model".

Refuted and unproved. Petit–Quisquater ($2^{c\ n^{2/3}\ \log\ n}$), Semaev 2015 (FIPS $n = 409, 571$ "theoretically broken") and Nagao 2013/549 all rest on $D_{reg} \approx d_{ff}$; the measured D_{reg} for S_3 with a random subspace runs $3, 4, 4, \geq 5$ at $n = 16, 20, 30, 40$ (Kosters–Yeo) and 5 at $n = 45$ for Semaev's own system (126 GB). FPPR's own extrapolation for $n = 131, m = 2$ is $2^{74.5}$ Gröbner operations, "still worse than exhaustive search" (Renault's EUROCRYPT 2012 slides). Galbraith (ECC 2015): prime $n > 160$ "seems to be completely immune to point decomposition attacks, even just trying to get a cube-root algorithm (case $m = 4$)". Yokoyama–Yasuda– Takahashi–Kogure (JMC 2020) prove the naive (non-descended) method cannot beat generic algorithms. ECC2K-130 is unsolved: the project page still shows the 2009 $2^{60.9}$-iteration estimate and no completion, Certicom lists the 131-bit challenges as open, and the largest binary ECDLP solved by any method is 113 bits (sect113r2, FPGA rho, 2016).

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5. Nagao's decomposition against summation polynomials

Verdict: no, and the reasons are structural rather than a matter of engineering.

1. **Same object, un-eliminated.** $S_{m+1}(x_1, \ldots, x_m, x_R) = 0$ is the condition that some $f \in L((m + 1)\cdot O)$ vanishes at $P_1, \ldots, P_m, -R$. Nagao keeps the coefficients of f as unknowns (quadratic, many variables); Semaev eliminates them (one polynomial, degree 2^{m-1} per variable, few variables). The Bézout degrees agree. Which point on that axis is best is Galbraith's open tradeoff problem, and every published point (§4.2) and every point measured here (§2) is enumeration-bound at the sizes reached.
2. **The gain Nagao offers needs a subfield.** "Coefficients in F_q" is a linear condition; "roots in V" is not. At $n = 131$ the method has no handle, and this repository's attempt to give it one — the function-first support test $H \mid L_V$ and its image-space successor — is the best evidence: it beats the Semaev encodings under the same SAT solver on the real curve ($32/32$ against $0/32$ at $d = 6$) and then loses to pair enumeration by $2.02\times$ in counted field operations, because it visits $\approx |F|^2$ candidates where the double loop visits $|F|^2/2$.
3. **Where Nagao-type relations do apply, the repository already has them.** The cover-and-decomposition ledger on F_{p^6} (genus-3 cover, Nagao relations, Joux–Vitse sieve reproduced at $\approx 10^3\times$ per relation) is the composite-degree case; its crossover with rho is measured at $\ell \approx 2^{50}$ on the weak class only. None of it transfers to prime n.
4. **The arity and base-shape limits are intrinsic.** The RR chart fixes $m = 3$ and needs an F_2-subspace base; a Hamming-weight base or more summands are outside the encoding, not merely untested.

What this leaves, from the solver panel's own §11: an encoding that is linear after a preprocessing cheaper than $|F|^2$ (a statement about the group law on V, not about Nagao), and the non-subspace base. Neither is a Nagao direction.

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6. What is not exhausted: ranked, with falsifiers

Each item names the harness it reuses, the measurement, the falsifier, the expected class, and the cost. None is expected to be an advance. Items are ordered by (value of the answer) / (cost to get it). Python is not an option for any of them (`AGENTS.md`, "Implementation language"); the ecc2k130 codegen path would need its native replacement first.

#	item	reuses	measurement and falsifier	expected class	cost
1	**XOR-learning CDCL (Xorricane / CDXCL) on the frozen S'_4 corpus, $\ell = 8\ldots11$** — the one SAT route with a proof system stronger than DPLL + Gaussian elimination (XLIN ≡ RES($\oplus$)), and the open question Trimoska's thesis §9.1 names (WDSat + learning)	upstream benchmark generator (`EC-Index-Calculus-Benchmarks`), `semaev_corpus.rs` parameters, ANF → XNF converter to write (Rust)	conflicts per **uniform** UNSAT target against `2^{3ℓ}/3!` (WDSat's attained bound) and against `T_enum`. *Falsifier for the §1 verdict:* $< 0.5 \cdot$ `T_enum` at two consecutive $\ell \geq 8$. *Abandon if* $\geq 0.5 \cdot$ `2^{3ℓ}/3!` at $\ell = 9$, i.e. no pruning above the last level	engineering at best; the §3 argument predicts "closed"	1 week
2	**WDSat `-x` completeness audit, then a matched SATIC run** on the 60-input regression	`research/index_calculus_baseline_20260914/regression/` (runner is Python and must be replaced natively first, `AGENTS.md`), `wdsat_oracle.rs`	reproduce the wrong UNSAT (`solver_16`, 1/16 cells at $n = 131$) on an instance under the 23 M-entry cap, bisect XG-ext vs plain, report upstream; then SATIC (Blomme et al. 2025) on the same 60 inputs. *Falsifier:* SATIC summed median conflicts $< 0.5\times$ WDSat symmetry-on with no input $> 1.1\times$	accounting (the audit); engineering (SATIC)	3–5 days after the native runner
3	**PB-XOR solver on the Hamming-weight normal-basis base** (LinPB: RoundingSat + Gauss–Jordan, Yang–Meel CP 2021; or CryptoMiniSat's BNN/cardinality constraints) — the only base where the solver family is structurally different from Gröbner (§3.3), and Frobenius-stable at prime n with no storage	ONB Karatsuba circuit from `ecc2k130/codegen` (Python; needs the Rust emitter first), `cnf.py`'s Sinz counters as the control	median solve and conflicts at $n = 11, 23$, $w = 3$, $m = 3$, planted and uniform. *Falsifier:* no instance at $n = 23$ decided where CMS decided none in 25 min; per-bit exponent not below the measured `2^{2.3}`	engineering; the product law is base-independent	1–2 weeks
4	**Bosphorus ANF preprocessing** in front of CMS and Kissat on the frozen S'_4 corpus	corpus + `semaev_sat.rs` ANF writer	conflicts at $\ell = 8$ uniform before/after. *Falsifier:* $< 0.5\times$. Expected: it rediscovers the trace fall and degree-2	accounting	1 day

#	item	reuses	measurement and falsifier	expected class	cost
			Macaulay rows the native path already injects		
5	**Complete the SMT ladder to `n = 23`** so `research/pdp_smt_oracle_20261008/` can be classed (its P4 needs `n = 23`)	`pdp_smt_oracle_bench` / `_score` (Rust, exists)	`--n 19 --targets 12`, `--n 23 --targets 8`, seed 20261010, isolated host	accounting	hours of compute
6	**Yield-law cross-check against thesis Table 7.7** (observed decomposition probability ≈ 2× the `2^{mℓ}/(m!·2^n)` heuristic at `n = 24`, e.g. `0.372` vs `0.167` at `ℓ = 8`), against this repository's 12-cell confirmation of `C(|F|, m)/#E`	`ic` yield tooling	recount on the thesis' `(n, ℓ)` with signed points and S_4 root multiplicity separated; a 2× that survives is a constant in the targets-per-relation box of the product law	accounting	hours
7	**GPU / bit-sliced pairs-and-solve leaf** (the FES-style enumeration the SAT literature never benchmarks against)	`gpu/fes/` kernels (compiled, cross-checked, not wired; no throughput measured), `s3_x_roots` one-word kernel	candidates per second per device at `ℓ = 12…16` vs one core. *Falsifier:* `< 50×` one core	engineering; moves toy-ladder reach (a PDP record at `ℓ ≥ 14` would be new), not the ratio	1–2 weeks

Not worth a round, with the reason: any SMT theory (no `GF(2^n)` solver exists and a native one would be Gröbner); ML-guided SAT (no ECC benchmarks, poor size generalisation); in-memory or GPU *SAT* hardware (simulated or generic instances); Frobenius-invariant subspaces at `n = 131` (do not exist); cube-and-conquer on the descended system (its natural cubes are the `m − 1` summand prefixes, which is pairs-and-solve under another name); Semaev's 2015 cubic model (first-fall assumption contradicted at `n = 45`, Kosters–Yeo).

6.1 Outcomes (2026-10-11)

The seven items were attempted in one cloud session; evidence and protocols are in `research/pdp_followups_20261010/` (PR #1650). Every outcome is a stage diagnostic and none moves a ratio to the floor.

#	item	outcome	class
1	XOR-learning CDCL	**blocked**: the Xorricane source is unreachable under the environment's network policy; protocol frozen in `blocked/PROTOCOLS.md`	pending
2	WDSat `-x` audit, SATIC	**blocked**: upstream sources unreachable, Python runner needs a native replacement first; protocol frozen	pending

#	item	outcome	class
3	Hamming-weight ONB base under SAT	**run** with a native emitter (`examples/hamming_nb_sat_panel.rs`): **run** with a native emitter (`examples/hamming_nb_sat_panel.rs`; 128 cells, 75 verified relations, zero disagreements with the exact oracle): at `n = 23, m = 3` neither CryptoMiniSat nor CaDiCaL decides any of 8 targets in 180 s (P2 fails) although the pair table decides each in at most 0.36 s; at `n = 23, m = 2` every proved refutation (`w = 2`) costs `117 … 174×` the `m − 1` floor in conflicts, and no refutation is proved at `w = 3`; PB-XOR arm blocked (see `hamming_nb_sat/RESULTS.md`)	boundary (off-the-shelf CDCL on this base at `m = 3`, `n = 23`, 180 s)
4	Bosphorus preprocessing	**blocked**: source unreachable; protocol frozen	pending
5	SMT ladder to `n = 23`	**done**: P1 pass, P2 fail, P3 fail (cvc5-bv1 found 5/12 to the native solver's 4/12 at `n = 19`, nothing refuted by any arm), P4 unscorable; at `n = 23` every arm exhausted 120 s on all 8 targets. cvc5 ran as 1.1.2 (amendment 3)	engineering candidate for the constant, by the registered rule
6	yield-law cross-check	**done**: thesis Table 7.7's `P` is `#sat/#unsat`; the exact census on $y^2 + xy = x^3 + x^2 + 1$ over `F_{2^24}` gives a satisfiable fraction `0.56–0.83×` the `2^{3ℓ}/(3!·2^n)` heuristic (curve) and `0.85–1.1×` (twist); the mean yield law holds to three digits; at $\lambda \approx 1.3$ the hit probability is `0.46` against Poisson's `0.72`. The thesis' `2×` is not reproduced and cannot come from the yield law	accounting
7	GPU pairs-and-solve leaf	**blocked** (no GPU); the CPU reference arm is measured to $\ell = 11$ (0.651 s per refuting target, `2^{21}` pairs)	pending

Two of the §6 falsifiers were testable and neither fired: item 5 found no SMT arm refuting a cell above `n = 13`, and item 3 found every proved refutation at `≥ 117×` the enumeration floor. The §1 verdict stands.

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7. Closed: do not re-run

- **A third SAT solver on the subspace Semaev system.** `solver_16` §7: "not more `d`, not more budget, and not a third SAT solver on the same descended system." Kissat, CaDiCaL, CryptoMiniSat, WDSat, the native solver, cvc5 and Z3 are all measured; they differ by constants around `T_enum · 2^ℓ/m`.
- **Riemann–Roch encodings under SAT.** `3.5·10⁴×` the S'_4 conflicts at `d = 5`; "the coefficient block hides the certificate from the solver."

- **Symmetry breaking inside the IC pipeline's SAT oracle.** Measured not to pay once the degree-2 Macaulay rows are present (5,512 → 6,412 conflicts at `n = 9, m = 3`); WDSat's own symmetry breaking is worth exactly the `3!` and no more.
- **One-hot Frobenius-phase variables in SAT.** `n = 19` fails at 1,200 s.
- **Crossbred as the pipeline oracle.** Loses to the double loop `3.4×–1800×`.
- **Hamming ideals under truncated Gröbner.** H0 stands at every size run.

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8. Visuals and graphs checked

- `research/pdp_sat_smt_survey_20261010/oracle_ladder.svg`: measured per-target cost of the SAT arms against the enumeration floor over `ℓ` (data from the rows of §2 that report conflicts).
- `research/pdp_sat_smt_survey_20261010/product_law.svg`: where a solver constant sits inside the `m·2^n/n` identity, and the `2^{−63}` it would need.
- `research/pdp_sat_smt_survey_20261010/REPORT.pdf`: this note with both figures, built by pandoc and headless Chromium (`build.sh`).
- Canonical graphs checked and left unchanged, because no operation count or verdict moved: `docs/index-calculus-scoreboard.html`, `docs/ic/progress-timeline.json`, `docs/ic/leaderboard.json`, `docs/ic/BOUNDARY_TARGETS.md`. The boundary target "sub-`2^{2ℓ}` oracle at fixed `ℓ = 8`" stays open and un-attempted by anything in §6.

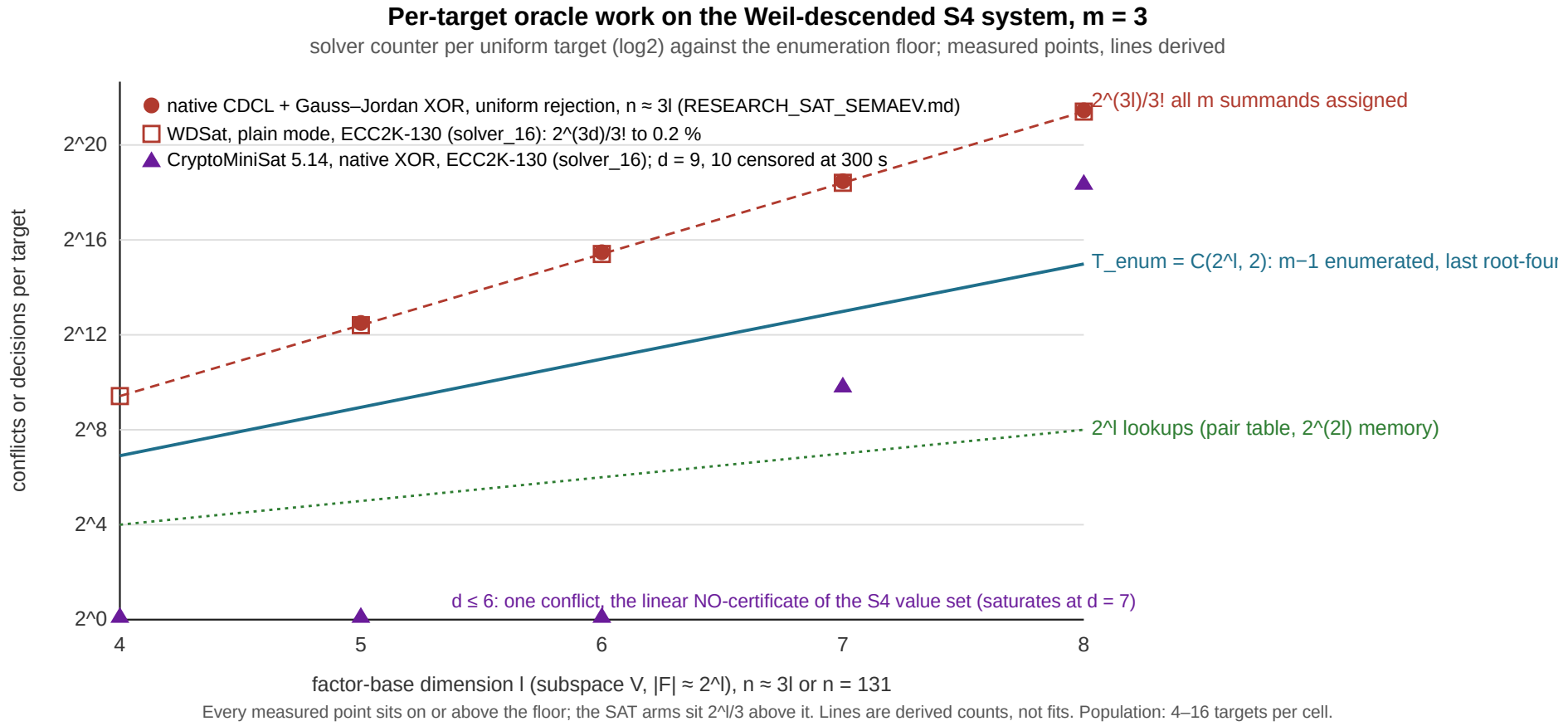
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Repository artefacts are linked inline. External sources, in the order the note uses them; items marked † were read as abstracts or secondary restatements only (ePrint and HAL were behind bot challenges from this environment).

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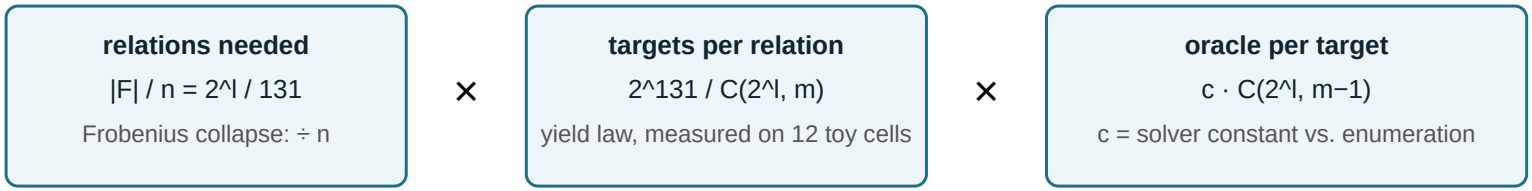
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Figures



Per-target oracle work against the enumeration floor

Where a solver constant sits in the index-calculus product law (ECC2K-130, Frobenius-stable base)



= c · m · 2^131 / 131 (every 2^l cancels: l, m and the base move work between boxes, never the total)

Pollard rho with ⟨−1⟩ × ⟨π⟩ (reference):	`2^60.81`
c needed to reach rho (m = 3):	`2^−64.8`
c measured for every SAT / SMT / WDSat arm in the tree:	`2^l / m ≥ 1`
c for pairs-and-solve, pair table, MITM:	`1 (2^−l with 2^(2l) memory)`

Sources: RESEARCH_ECC2K130_DECOMPOSITION.md §5.1–5.3 and §6 (derived), the §2 table of this note (measured). Nothing here is a run.

Where a solver constant sits in the product law